



Combined effects of particulate and gaseous pollutants on Alzheimer's disease-related brain imaging markers and cognitive function[☆]

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ABSTRACT

The combined effects of various air pollutants on Alzheimer's disease (AD)-specific brain structural pathologies are poorly understood, and whether these effects lead to impairments in specific cognitive domains has not been determined. We examined the associations between exposure to air pollutant mixtures and subclinical AD-related brain imaging markers, and then assessed whether AD-specific cortical atrophy mediated the association between air pollution exposure and specific cognitive domains. This cross-sectional study included 649 adults without dementia (≥ 50 years) residing in 2 metropolitan and 2 small cities in the Republic of Korea, all of whom underwent brain magnetic resonance imaging scan and neuropsychological testing. The 5-year average concentrations of particulate and gaseous pollutants were estimated at participants' residential addresses. To estimate the combined effects of these pollutant mixtures, we applied Bayesian kernel machine regression (BKMR) to capture potential non-linearity and interaction effects, and quantile g-computation (QGcomp) to estimate overall mixture effects. We additionally performed partial least squares structural equation modeling to test mediation pathways via brain cortical atrophy in the association between air pollutant mixtures and cognitive domains. BKMR analysis indicated significant associations between air pollutant mixtures and AD-related imaging markers, while QGcomp analysis revealed associations with reduced whole-brain cortical thickness. NO₂ exhibited the most prominent effects on subclinical imaging markers. AD-specific cortical atrophy fully mediated the associations of air pollutant mixtures with language, visual memory, and frontal/executive functions, and partially mediated those with visuospatial function, with the greatest mediation effect observed for visuospatial function, followed by language, visual memory, and frontal/executive function. These findings suggest that exposure to air pollutant mixtures may be associated with AD-specific cortical atrophy and a subsequent decline in cognitive functions in dementia-free adults.

[☆] This paper has been recommended for acceptance by.

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1. Introduction

The brain toxicity of ambient air pollutants has been investigated in numerous epidemiological studies. Three meta-analyses of these epidemiological studies concluded that exposure to particulate matter (PM) increases the risk of neurodegenerative disorders, such as Alzheimer's disease (AD) (Fu et al., 2019; Tsai et al., 2019; Gong et al., 2023). Neuroimaging studies have corroborated such harmful effects of PM by demonstrating reductions in cortical volume and thickness (subclinical imaging phenotypes of AD) associated with PM exposure (Gale et al., 2020). Some previous studies have also demonstrated that exposure to nitrogen dioxide (NO₂), a traffic-related gaseous pollutant, is inversely associated with cortical volume or thickness (Cho et al., 2020; Crous-Bou et al., 2020; Gale et al., 2020). However, these previous studies have used single- or two-pollutant models, even though natural exposure to air pollution involves the simultaneous inhalation of multiple pollutants. Therefore, it is now warranted to examine the combined effects of air pollutant mixtures on both subclinical and clinical phenotypes of AD.

Compared with PM and NO₂, airborne heavy metals and volatile organic compounds (VOCs) have not been exhaustively examined in studies linking air pollution to AD. Additionally, many toxicological studies have suggested the effects of heavy metals, such as manganese, on subclinical and clinical phenotypes of AD (e.g., increased amyloid beta peptides, impaired working memory) (Schneider et al., 2006; Liapi et al., 2008; Lin et al., 2020). However, human studies linking blood manganese levels to AD phenotypes have yielded conflicting findings (Du et al., 2017); this might be attributed to human studies using blood manganese as the indicator of exposure to manganese, an essential mineral in small amounts. One of our existing studies focused on inhalation exposure to airborne manganese and found an inverse association with brain cortical thickness in the parietal and occipital lobes (Woo et al., 2023). However, limited evidence has suggested an effect of manganese inhalation exposure on the brain. In *in vivo* studies, VOCs have also been suggested to act as neurotoxins that induce oxidative stress (Wang et al., 2014; Kodavanti et al., 2015; Wang et al., 2018). Nevertheless, epidemiological evidence on the effects of VOCs on subclinical imaging phenotypes of AD is lacking.

Therefore, in the present study, we investigated the combined effects of inhalation exposure to PM, NO₂, manganese, and VOCs on subclinical imaging markers (whole-brain cortical thickness and AD-specific cortical atrophy) in adults without dementia. Here, we used Bayesian kernel machine regression (BKMR) and quantile g-computation (QGcomp), analytical approaches that better reflect real-world exposure scenarios where humans are simultaneously exposed to multiple pollutants. These methods allow us to estimate overall mixture effects while accounting for potential non-linear relationships and interactions among pollutants (Bobb et al., 2015; Keil et al., 2020). Furthermore, we examined whether the combined effects of the air pollutant mixtures on the subclinical imaging markers impair specific domains of cognitive function (frontal/executive function, language, visuospatial function, verbal memory, and visual memory), using partial least squares structural equation modeling (PLS-SEM).

2. Methods

2.1. Study participants

The Environmental Pollution-Induced Neurological Effects (EPINEF) study, initiated in 2014, is a multicenter community-based cohort study focusing on the neurological impacts of environmental pollutants in adults (Jang et al., 2021). In this study, we recruited participants aged 50 years and older from the Republic of Korea via local advertisements in four cities: Seoul, Incheon, Wonju, and Pyeongchang. The exclusion criterion was individuals with a history of neurological diseases (dementia, stroke, or Parkinson's disease). In the EPINEF study, from 2014

to 2019, 3775 participants enrolled and underwent baseline examinations at 3 university-based hospitals: Yonsei University Severance Hospital, Gachon University Gil Medical Center, and Wonju Severance Christian Hospital. In the EPINEF study, self-reported questionnaire data encompassing demographics, lifestyle behaviors, and health information were collected, along with anthropometric and blood pressure measurements. Blood samples were obtained after a minimum 12-h fasting period and analyzed at a central laboratory (Seoul Clinical Laboratory Co. Ltd, Seoul, Republic of Korea). A proportion of the participants underwent neuroimaging scans and neuropsychological testing. Neuroimaging scanning was conducted using a Philips 3T Achieva magnetic resonance imaging (MRI) scanner to assess brain structures. Simultaneously, neuropsychological testing was performed using the Seoul Neuropsychological Screening Battery, 2nd edition, to evaluate specific cognitive function domains (Jahng et al., 2015). A total of 649 participants completed both the neuroimaging scans and the neuropsychological testing with no missing data.

We obtained informed consent from participants before enrollment, and all procedures followed the ethical guidelines of the Declaration of Helsinki and relevant national and institutional committees on human experimentation. The present study was approved by the Yonsei University Health System Institutional Review Board (approval no. 4-2022-0418).

2.2. Exposure assessment

We computed long-term average concentrations of the following ambient air pollutants at each participant's residential address: PM with aerodynamic diameters of $\leq 10 \mu\text{m}$ (PM₁₀) and $\leq 2.5 \mu\text{m}$ (PM_{2.5}), NO₂, manganese in PM₁₀ (Mn₁₀), and VOCs (acetaldehyde, styrene, and toluene). For PM₁₀, NO₂, and PM_{2.5}, we predicted annual average concentrations using a universal kriging model developed based on regulatory monitoring data for the years 2001–2016 and 2015, as described elsewhere (Kim and Song, 2017). We computed the 5-year average concentrations of PM₁₀ and NO₂ by using these annual average concentrations. We used the 1-year average concentration of PM_{2.5} for 2015 when nationwide regulatory monitoring first became available. The performance of these prediction models was assessed using cross-validation, yielding R^2 values of 0.45 and 0.82 for PM₁₀ and NO₂, respectively, and 0.43 for PM_{2.5} in 2015. For the other pollutants (Mn₁₀, acetaldehyde, styrene, and toluene), we applied a universal kriging model to seasonal average concentrations, using measurements from the EPINEF monitoring campaign conducted from fall 2016 to summer 2017 for a subset of the EPINEF study participants. Model performance was evaluated using cross-validation R^2 values, and the respective values for four seasons were as follows: 0.31–0.85 for Mn₁₀, 0.25–0.68 for acetaldehyde, 0.23–0.67 for styrene, and 0.26–0.51 for toluene.

2.3. Assessments of subclinical and clinical phenotypes related to AD

The AD-related phenotypes included subclinical imaging (AD-specific cortical atrophy score and whole-brain cortical thickness) and clinical markers (specific cognitive function domains). A 3D T1 magnetization-prepared rapid gradient-echo image was obtained using a standardized protocol to extract the aforementioned subclinical imaging markers. The markers were then estimated using an automated brain image analysis tool (Inbrain®; MIDAS Information Technology Co., Ltd., Seongnam, Republic of Korea). Whole-brain cortical thickness was determined by averaging the cortical thickness of each mesh vertex. The AD-specific cortical atrophy score quantified the similarity of cortical atrophy patterns to those observed in patients with AD, as described in previous studies, with an accuracy of 91 % (Lee et al., 2018; Cho et al., 2023). A higher AD-specific cortical atrophy score indicates a greater resemblance to the typical cortical atrophy pattern observed in patients with AD. Specific cognitive function domains were assessed using the Seoul Neuropsychological Screening Battery, 2nd edition, a

standardized, comprehensive cognitive test widely used in the Republic of Korea (Ryu and Yang, 2023). We assessed the five domains: frontal/executive function, language, verbal memory, visual memory, and visuospatial function. Frontal/executive functions were evaluated using the Controlled Oral Word Association Test phonemic, Digit Symbol Coding, and Korean-Color Word Stroop Test. Language function was assessed using the Korean-Boston Naming Test, while visuospatial function was evaluated using the Rey Complex Figure Test (copy). Verbal memory was assessed using the Seoul Verbal Learning Test (delayed recall), and visual memory was evaluated using the Rey Complex Figure Test (delayed recall). Higher scores in each domain indicate superior cognitive performance.

2.4. Potential confounders

The potential confounders included in this study were age, sex, socioeconomic status (i.e., years of education), lifestyle habits (smoking status categorized as current, former, or never smoker; alcohol consumption categorized as current, former, or never drinker; physical activity categorized as vigorous, moderate, or inactive based on metabolic equivalent hours per week [MET-h/wk]), and intracranial volume derived from brain MRI scans (Ainsworth et al., 2000). The detailed definition of physical activity has been previously published (Park et al., 2024). Additionally, we considered the disease history, including diabetes mellitus, dyslipidemia, and hypertension. Diabetes mellitus was defined as a self-reported history of diabetes mellitus and a fasting blood glucose level ≥ 126.0 mg/dL or a hemoglobin A1C level ≥ 6.5 %. Dyslipidemia was defined as a self-reported history of dyslipidemia plus any of the following: total cholesterol ≥ 240.0 mg/dL, low-density lipoprotein cholesterol ≥ 160.0 mg/dL, high-density lipoprotein cholesterol < 40.0 mg/dL, or triglyceride ≥ 240.0 mg/dL. Hypertension was defined as a self-reported history of hypertension plus systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg. Body mass index (BMI) was calculated as weight in kilograms divided by the square of height in meters.

2.5. Statistical analysis

Continuous variables are presented as mean or median, along with their standard deviations (SDs) or the 25th to the 75th percentiles, respectively. Categorical variables are reported as frequencies and percentages (%). We used Spearman correlation analysis to assess the correlations among all pollutants. We standardized the levels of each air pollutant, each subclinical imaging marker (AD-specific cortical atrophy score and whole-brain cortical thickness), and each continuous covariate to a mean of 0 and variance of 1, expressing them as z-scores.

After adjusting for potential confounders, we conducted a linear regression analysis to explore the effects of exposure to a single air pollutant on subclinical imaging markers. All models included age, sex, BMI, years of education, cohabitation status, lifestyle habits (smoking status, alcohol consumption, and physical activity), and history of diseases (diabetes mellitus, dyslipidemia, and hypertension). Age and education level were excluded from the AD-specific cortical atrophy score analysis because the score was adjusted for these variables. Intracranial volume was included in the model to regress against whole-brain cortical thickness. The results are reported as the change (β) in subclinical imaging markers and a 95 % confidence interval per unit increase in the SD of air pollutant levels.

Considering potential non-linear interactions among air pollutant mixtures, we implemented the BKMR model to estimate the combined effects of multiple pollutants on subclinical imaging markers (Bobb et al., 2015). The BKMR model, a non-parametric approach, enables exploration of the combined effects of multiple pollutants on health outcomes, considering non-linear interactions between exposures. We fitted this model by using a Markov chain Monte Carlo algorithm with 10,000 iterations and a Gaussian kernel function to calculate each

pollutant's posterior inclusion probability. The combined effects of multiple pollutants on subclinical imaging markers were assessed by comparing the dose-response function values at specific percentiles (ranging from the 25th to the 75th percentile) to the median value (50th percentile), which served as the reference. Dose-response relationships between the seven air pollutants and subclinical imaging markers were analyzed by fixing the exposure levels of all other pollutants at their median values.

Furthermore, we examined the individual effect of a single pollutant by comparing the risk associated with the 75th percentile of a particular pollutant to its 25th percentile while maintaining all other pollutants fixed at their 25th, 50th, and 75th percentiles. We also evaluated the effects of air pollutant mixtures using a QGcomp model, a novel approach that combines weighted quantile sum regression with basic g-computation (Keil et al., 2020). We estimated the effects of air pollutant mixtures on subclinical imaging markers when all standardized exposures were concurrently increased by one quantile, regardless of the correlation direction between the air pollutant mixtures and subclinical imaging markers. When air pollutants exhibit varying directional effects, the QGcomp model assigns positive or negative weights to each pollutant, totaling either 1 or -1.

To test the path from exposure to air pollutant mixtures to specific cognitive function domains via brain cortical atrophy, we conducted PLS-SEM (Sarstedt et al., 2021) which is advantageous as it does not require normality assumptions and can accommodate latent variables comprising both formative and reflective measurement models. We considered air pollutants as exposures, specific cognitive function domains as outcomes, and subclinical imaging markers as mediators (Fig. 1A and B). In this model, air pollutants are described by formative measurement models, while reflective measurement models describe frontal/executive cognitive functions. Due to the high correlation between air pollutants (eFig. 1), we employed principal component analysis with varimax rotation to reduce dimensionality. The scree plot and interpretability determined the number of factors (den Reijer et al., 2021). Upon examining the factors, we found that PM_{2.5} did not significantly impact any of the outcomes or mediators. Therefore, we identified three factors that accounted for 96 % of the total variance after excluding PM_{2.5}. Factor 1 included styrene, Mn₁₀, and toluene; Factor 2 included NO₂ and acetaldehyde; Factor 3 consisted of PM₁₀ (eTable 1). We evaluated formative and reflective measurement models to verify the PLS-SEM by confirming the standardized indicators and their cut-off values suggested in previous studies (Hair et al., 2019; Sarstedt et al., 2021). We conducted separate PLS-SEM for each mediator to account for correlation and directionality between the mediators. Direct and indirect (specific-indirect) effects were estimated by evaluating the path coefficient and 95 % confidence interval with a 5000-sample bootstrapping procedure. To avoid missing some important paths, we did not perform corrections for multiple comparisons.

All statistical analyses were conducted using SAS software version 9.4 (SAS Institute, Cary, NC, USA). We used the *bkmr* and the *qgcomp* packages for BKMR and QGcomp analysis, respectively, in R statistical software (version 4.1.1; R Foundation for Statistical Computing, Vienna, Austria). PLS-SEM analysis was performed using SmartPLS (version 4). Two-sided *p*-values below 0.05 were considered statistically significant.

3. Results

3.1. Characteristics of study participants

The mean (SD) age of the 649 participants was 67.1 (6.8), and the number of women was 348 (53.6 %) (Table 1). Among 649 participants, 5.7 % (*n* = 37) were current smokers, and 94.3 % (*n* = 612) were non-smokers. The median values and interquartile ranges (IQR) for the seven air pollutants were as follows: PM₁₀, 50.4 (47.3, 53.1) $\mu\text{g}/\text{m}^3$; PM_{2.5}, 25.7 (25.3, 26.2) $\mu\text{g}/\text{m}^3$; NO₂, 31.6 (16.6, 35.4) ppb; Mn₁₀, 12.3 (9.6, 14.1) $\mu\text{g}/\text{m}^3$; acetaldehyde, 11.1 (9.4, 11.7) $\mu\text{g}/\text{m}^3$; styrene, 16.4 (12.7,

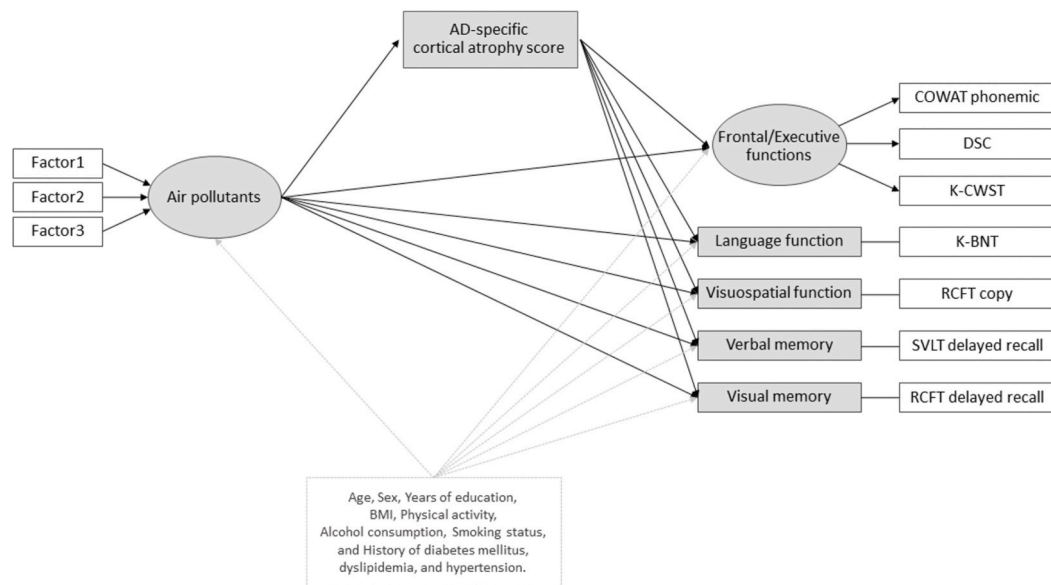
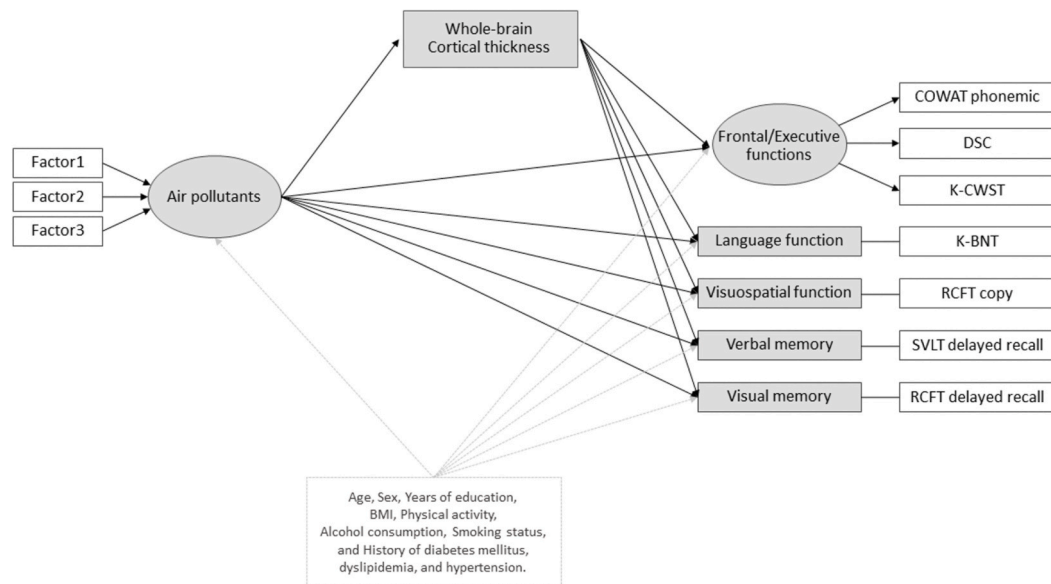
(A) AD-specific cortical atrophy score**(B) Whole-brain cortical thickness**

Fig. 1. Structural model of partial least squares structural equation modeling (PLS-SEM) for mediation analysis to estimate the indirect effects of air pollution exposure on specific cognitive function domains mediated by AD-specific cortical atrophy score (A) or whole-brain cortical thickness (B). AD, Alzheimer's disease; COWAT, Controlled Oral Word Association Test; DSC, Digit Symbol Coding; K-CWST, Korean-Color Word Stroop Test; K-BNT, Korean version of the Boston Naming Test; RCFT, Rey Complex Figure Test; SVLT, Seoul Verbal Learning Test; BMI, body mass index. **Notes.** The rectangles indicate the observed variables, and the ellipses indicate the latent variables/constructs. Factor 1 includes styrene, manganese particles with an aerodynamic diameter $\leq 10 \mu\text{m}$ (Mn_{10}), and toluene. Factor 2 comprises nitrogen dioxide (NO_2) and acetaldehyde. Factor 3 consists only of particulate matter with an aerodynamic diameter $\leq 10 \mu\text{m}$ (PM_{10}). All categorical variables were converted to dummy variables.

19.7) $\mu\text{g}/\text{m}^3$; and toluene, 33.7 (27.8, 38.0) $\mu\text{g}/\text{m}^3$. The median (IQR) of the AD-specific cortical atrophy score and whole-brain cortical thickness were 13.5 (3.9, 43.3) and 2.3 (2.3, 2.4) mm, respectively.

3.2. Correlations between air pollutants

We found that NO_2 , Mn_{10} , acetaldehyde, styrene, and toluene were moderately to highly correlated with each other (Spearman's $\rho =$

0.68–0.94). PM_{10} and $\text{PM}_{2.5}$ were weakly to moderately correlated with each other (Spearman's $\rho = 0.25$ –0.56) (eFig. 1). All correlations were statistically significant ($p < 0.001$).

3.3. Associations between exposure to a single air pollutant and subclinical imaging markers

In the linear regression model, a 1-unit SD increase in NO_2 was

Table 1
Characteristics of study participants.

Characteristics	Total (N = 649)
Age (years), mean (SD)	67.1 (6.8)
Sex, N (%)	
Men	301 (46.4)
Women	348 (53.6)
Years of education (years), mean (SD)	10.4 (4.2)
Body mass index (kg/m ²), mean (SD)	24.6 (2.9)
Smoking status, N (%)	
Current smoker	37 (5.7)
Former smoker	188 (29.0)
Non-smoker	424 (65.3)
Alcohol consumption, N (%)	
Current drinker	276 (42.5)
Non-drinker	373 (57.5)
Physical activity, N (%)	
Vigorous	84 (12.9)
Moderate	330 (50.9)
Inactive	235 (36.2)
History of diseases, N (%)	
Diabetes mellitus	138 (21.3)
Dyslipidemia	349 (53.8)
Hypertension	199 (30.7)
Ambient air pollutants, median (IQR)	
PM ₁₀ , µm/m ³	50.4 (47.3, 53.1)
PM _{2.5} , µm/m ³	25.7 (25.3, 26.2)
NO ₂ , ppb	31.6 (16.6, 35.4)
Mn ₁₀ , µm/m ³	12.3 (9.6, 14.1)
Acetaldehyde, µm/m ³	11.1 (9.4, 11.7)
Styrene, µm/m ³	16.4 (12.7, 19.7)
Toluene, µm/m ³	33.7 (27.8, 38.0)
AD-specific cortical atrophy score, median (IQR)	13.5 (3.9, 43.3)
Whole-brain cortical thickness (mm), median (IQR)	2.3 (2.3, 2.4)
Intracranial volume (1000 mm ³), mean (SD)	1523.0 (169.5)

Abbreviations: SD: Standard deviation; IQR: interquartile range; PM₁₀: particulate matter, aerodynamic diameter ≤10 µm; PM_{2.5}: particulate matter, aerodynamic diameter ≤2.5 µm; NO₂: nitrogen dioxide; Mn₁₀: manganese in PM₁₀; AD: Alzheimer’s disease.

significantly associated with an increased AD-specific cortical atrophy score ($\beta = 0.18$, 95 % confidence interval [CI] = 0.10, 0.26, $p < 0.001$) (Fig. 2). A 1-unit SD increase in acetaldehyde was significantly associated with an increased AD-specific cortical atrophy score ($\beta = 0.11$, 95 % CI = 0.03, 0.20, $p = 0.01$). Exposure to each air pollutant was significantly associated with reduced whole-brain cortical thickness,

with β values ranging from -0.45 to -0.09 (all p values < 0.05).

3.4. Associations between air pollutant mixtures and subclinical imaging markers

In the BKMR model, the conditional posterior inclusion probabilities for NO₂ were highest (1.0), and those for PM_{2.5} were lowest (0.302 for AD-specific cortical atrophy score; 0.405 for whole-brain cortical thickness) (eTable 2). None of the dose-response relationships between each air pollutant and subclinical imaging markers displayed prominent non-linearity, except for the relationship between NO₂ and whole-brain cortical thickness (eFig. 2). Overall, mixture effect visualization showed that the AD-specific cortical atrophy score was significantly lower at the 25th percentile than at the 50th percentile of all air pollutants (Fig. 3A). Whole-brain cortical thickness was significantly greater at all air pollutants’ 25th percentiles than at the 50th percentiles (Fig. 3B). In visualizing interactions, the positive association between NO₂ and AD-specific cortical atrophy score tended to be stronger at the 25th percentiles than at the 50th and 75th percentiles of all the other pollutants (eFig. 3A). By contrast, the associations between toluene and AD-specific cortical atrophy scores tended to be weaker at the 25th percentile than at the 50th and 75th percentiles of all the other pollutants. Similar patterns were observed in the associations with the whole-brain cortical thickness (eFig. 3B). These findings indicate possible interactions between the air pollutants.

In the QGcomp model, increasing all seven air pollutant mixtures concentrations by one quartile was significantly associated with reduced whole-brain cortical thickness ($\beta = -0.30$, 95 % CI = -0.39 , -0.22). However, the association with increased AD-specific cortical atrophy score did not reach statistical significance ($\beta = 0.08$, 95 % CI = -0.02 , 0.17) (Table 2). NO₂ exhibited the largest positive weight (0.60) in the mixture effect on the AD-specific cortical atrophy score and the largest negative weight (-0.53) on whole-brain cortical thickness (Table 2; eFig. 4).

3.5. Mediating role of brain cortical atrophy in the association between air pollutant mixtures and specific domains of cognitive function

The PLS-SEM demonstrated a significant direct path from air pollutant mixtures to AD-specific cortical atrophy ($p < 0.001$) (Table 3). The AD-specific cortical atrophy score was significantly associated with

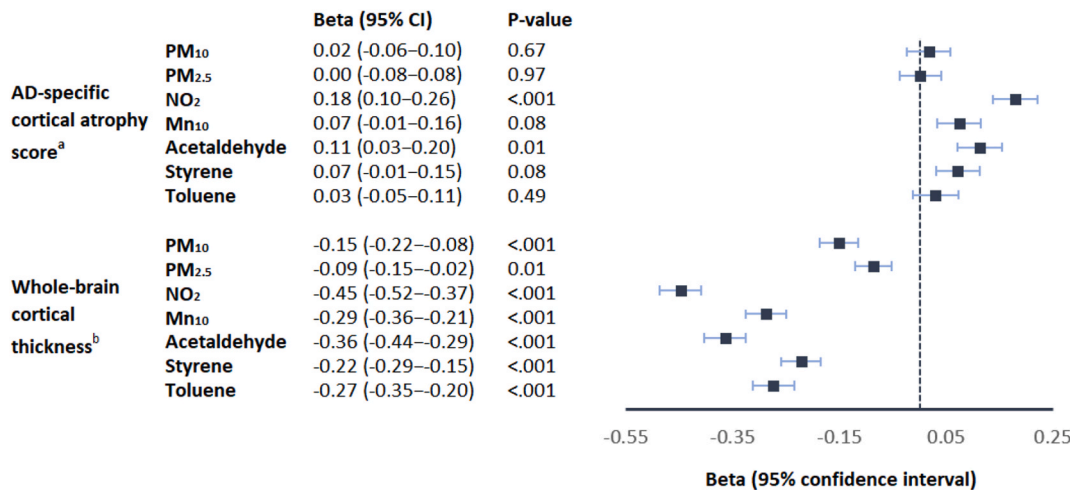


Fig. 2. Associations between exposure to a single air pollutant and subclinical imaging markers. CI, confidence interval; AD, Alzheimer’s disease; PM₁₀, particulate matter with an aerodynamic diameter ≤10 µm; PM_{2.5}, particulate matter with an aerodynamic diameter ≤2.5 µm; NO₂, nitrogen dioxide; Mn, manganese. **Notes.** Beta coefficients (per 1 standard deviation in each pollutant) were estimated from linear regression analyses. ^aModel was adjusted for sex, body mass index, physical activity, alcohol consumption, smoking status, and history of diabetes mellitus, dyslipidemia, and hypertension. ^bModel was adjusted for age, sex, years of education, body mass index, physical activity, alcohol consumption, smoking status, intracranial volume, and history of diabetes mellitus, dyslipidemia, and hypertension.

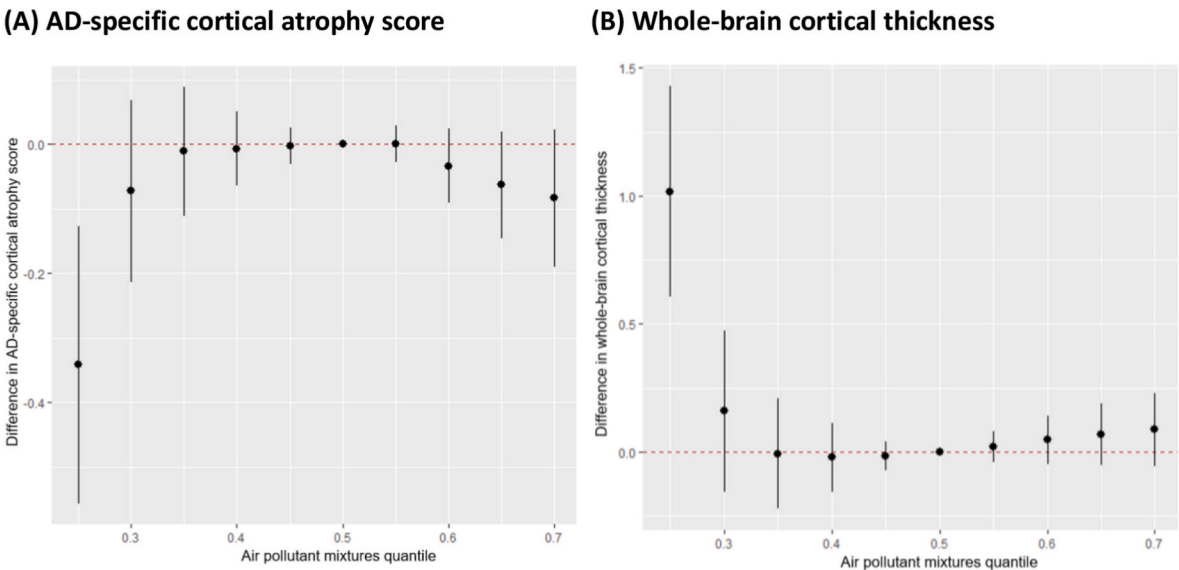


Fig. 3. Associations of air pollutant mixtures with AD-specific cortical atrophy score and whole-brain cortical thickness, estimated via Bayesian kernel machine regression analyses. AD, Alzheimer’s disease. **Notes.** The plot shows the overall effect of the mixture and 95 % confidence intervals when all air pollutant levels are fixed at a certain percentile compared with the case when all air pollutant levels are fixed at the 50th percentile. Model (A) was adjusted for sex, body mass index, physical activity, alcohol consumption, smoking status, and history of diabetes mellitus, dyslipidemia, and hypertension. Model (B) was adjusted for the model (A) variables plus age, years of education, and intracranial volume.

Table 2
Associations of air pollutant mixtures with AD-specific cortical atrophy score and whole-brain cortical thickness, estimated from quantile-based g-computation model.

Subclinical imaging markers	Exposures	Weight ^c	All participants (N = 649)
			Beta (95 % CI) ^d
AD-specific cortical atrophy score ^a	Air pollutant mixture	–	0.08 (–0.02, 0.17)
	PM ₁₀	0.05	–0.02 (–0.11, 0.07)
	PM _{2.5}	–0.04	0.01 (–0.07, 0.09)
	NO ₂	0.60	0.16 (0.05, 0.27)
	Mn ₁₀	0.12	0.06 (–0.13, 0.25)
	Acetaldehyde	0.09	0.02 (–0.12, 0.16)
	Styrene	0.15	0.03 (–0.11, 0.18)
	Toluene	–0.96	–0.20 (–0.32, –0.08)
Whole-brain cortical thickness ^b	Air pollutant mixture	–	–0.30 (–0.39, –0.22)
	PM ₁₀	–0.05	0.04 (–0.04, 0.12)
	PM _{2.5}	0.32	0.02 (–0.05, 0.09)
	NO ₂	–0.53	–0.19 (–0.29, –0.10)
	Mn ₁₀	0.04	–0.06 (–0.22, 0.10)
	Acetaldehyde	–0.42	–0.19 (–0.32, –0.07)
	Styrene	0.41	0.12 (–0.004, 0.24)
	Toluene	0.23	–0.03 (–0.13, 0.07)

Abbreviations: CI, confidence interval; AD, Alzheimer’s disease; PM₁₀, particulate matter ≤10 μm in aerodynamic diameter; PM_{2.5}, particulate matter ≤2.5 μm in aerodynamic diameter; NO₂, nitrogen dioxide; Mn₁₀, manganese in PM₁₀. **Notes.** ^aModel adjusted for sex, body mass index, physical activity, alcohol consumption, smoking status, and history of diabetes mellitus, dyslipidemia, and hypertension. ^bModel adjusted for age, sex, years of education, body mass index, physical activity, alcohol consumption, smoking status, intracranial volume, and history of diabetes mellitus, dyslipidemia, and hypertension. ^cWeights indicate the contribution of each air pollutant to the association between air pollutant mixtures and subclinical imaging markers. ^dBeta coefficients per one-quantile increase in standardized air pollutants or air pollutant mixtures.

decreased frontal/executive functions, language, visual memory, and visuospatial functions. Air pollutant mixtures negatively and indirectly affected specific cognitive function domains via AD-specific cortical atrophy. Air pollutant mixtures were associated with decreases in frontal/executive function ($\beta = -0.01$, 95 % bias-corrected [BC] CI = $-0.03, -0.003$), language function ($\beta = -0.02$, 95 % BC CI = $-0.04, -0.01$), visual memory function ($\beta = -0.02$, 95 % BC CI = $-0.04, -0.004$), and visuospatial function ($\beta = -0.03$, 95 % BC CI = $-0.05, -0.01$) via AD-specific cortical atrophy. Considering the significance of both direct and indirect pathways, we found that AD-specific cortical atrophy fully mediated the associations between air pollutant mixtures and decreased language, visual memory, and frontal/executive functions, partially mediated the associations between air pollutant mixtures and decreased visuospatial function. The magnitude of mediation effects was greatest for visuospatial function, followed by language, visual memory, and frontal/executive function. Contrastingly, air pollutant mixtures had no significant indirect effects on the specific cognitive function domains via whole-brain cortical thickness. All indicators in PLS-SEM fulfilled the standardized criteria (eTable 3).

4. Discussion

To the best of our knowledge, this study is the first to use state-of-the-art techniques to investigate the combined effects of particle and gaseous pollutants on brain cortical atrophy in individuals without dementia. We also examined whether brain cortical atrophy attributable to air pollutant mixtures impairs specific cognitive function domains. The BKM analysis demonstrated that air pollutant mixtures significantly affected brain cortical thickness, particularly in an AD-specific manner. Among the studied pollutants, NO₂ was a major contributor to the effects of air pollutant mixtures on brain cortical atrophy. Moreover, PLS-SEM demonstrated that the effects of air pollutant mixtures on frontal/executive functions, language, and visual memory were fully mediated by AD-specific brain cortical atrophy. Notably, AD-specific cortical atrophy attributable to air pollutant mixtures may be more critical to the deterioration of various cognitive functions than global cortical atrophy.

Numerous epidemiological studies have investigated the associations between PM and NO₂ and brain MRI markers (Chen et al., 2015; Power et al., 2018; Cho et al., 2020). However, evidence on the associations

Table 3
Effects of air pollutant mixtures on specific cognitive function domains via subclinical imaging markers.

Mediator	Paths	Path coefficient (95 % CI)	SE	t value	p-value	Decision
AD-specific cortical atrophy score (ADCAS)	<i>Direct effect</i>					
	AP → ADCAS	0.18 (0.09, 0.27)	0.05	3.94	<0.001	Supported
	AP → Frontal/executive functions	0.05 (−0.02, 0.12)	0.04	1.50	0.13	Not supported
	AP → Language function	−0.06 (−0.14, 0.02)	0.04	1.49	0.14	Not supported
	AP → Verbal memory	−0.003 (−0.09, 0.08)	0.04	0.06	0.95	Not supported
	AP → Visual memory	−0.08 (−0.17, 0.005)	0.04	1.93	0.05	Not supported
	AP → Visuospatial function	0.18 (0.11, 0.26)	0.04	4.67	<0.001	Supported
	ADCAS → Frontal/executive functions	−0.08 (−0.14, −0.01)	0.03	2.49	0.01	Supported
	ADCAS → Language function	−0.11 (−0.19, −0.03)	0.04	2.58	0.01	Supported
	ADCAS → Verbal memory	−0.06 (−0.13, 0.01)	0.04	1.60	0.11	Not supported
	ADCAS → Visual memory	−0.10 (−0.18, −0.02)	0.04	2.46	0.01	Supported
	ADCAS → Visuospatial function	−0.15 (−0.24, −0.08)	0.04	3.66	<0.001	Supported
	<i>Indirect effect</i>	Coefficient point estimate (95 % BC CI)				Decision
	AP → ADCAS → Frontal/executive functions	−0.01 (−0.03, −0.003)				Full mediation
	AP → ADCAS → Language function	−0.02 (−0.04, −0.01)				Full mediation
	AP → ADCAS → Verbal memory	−0.01 (−0.03, 0.001)				No mediation
	AP → ADCAS → Visual memory	−0.02 (−0.04, −0.004)				Full mediation
	AP → ADCAS → Visuospatial function	−0.03 (−0.05, −0.01)				Partial mediation
Whole-brain cortical thickness (CTh)	<i>Direct effect</i>					Decision
	AP → Whole-brain CTh	−0.49 (−0.55, −0.44)	0.03	17.59	<0.001	Supported
	AP → Frontal/executive functions	0.06 (−0.02, 0.14)	0.04	1.60	0.11	Not supported
	AP → Language function	−0.04 (−0.13, 0.05)	0.04	1.00	0.32	Not supported
	AP → Verbal memory	−0.01 (−0.10, 0.08)	0.05	0.21	0.83	Not supported
	AP → Visual memory	−0.08 (−0.17, 0.02)	0.05	1.58	0.11	Not supported
	AP → Visuospatial function	0.19 (0.12, 0.28)	0.04	4.63	<0.001	Supported
	Whole-brain CTh → Frontal/executive functions	0.04 (−0.04, 0.11)	0.04	1.03	0.30	Not supported
	Whole-brain CTh → Language function	0.06 (−0.03, 0.15)	0.05	1.33	0.18	Not supported
	Whole-brain CTh → Verbal memory	−0.01 (−0.09, 0.08)	0.04	0.12	0.91	Not supported
	Whole-brain CTh → Visual memory	0.04 (−0.05, 0.13)	0.04	0.91	0.36	Not supported
	Whole-brain CTh → Visuospatial function	0.06 (−0.01, 0.12)	0.03	1.65	0.10	Not supported
	<i>Indirect effect</i>	Coefficient point estimate (95 % BC CI)				Decision
	AP → Whole-brain CTh → Frontal/executive functions	−0.02 (−0.06, 0.02)				No mediation
	AP → Whole-brain CTh → Language function	−0.03 (−0.07, 0.01)				No mediation
	AP → Whole-brain CTh → Verbal memory	0.002 (−0.04, 0.04)				No mediation
	AP → Whole-brain CTh → Visual memory	−0.02 (−0.06, 0.02)				No mediation
	AP → Whole-brain CTh → Visuospatial function	−0.03 (−0.06, 0.004)				No mediation

Abbreviations: AD, Alzheimer’s disease; CI, confidence interval; SE, standard error; AP, air pollution; ADCAS, AD-specific cortical atrophy score; BC, bias-corrected; CTh, cortical thickness. **Notes.** Coefficients and 95 % confidence intervals were estimated from partial least squares structural equation modeling with bootstrap re-sampling (5000 iterations). All models were adjusted for sex (men and women), age, years of education, body mass index (categorized as ≥ 25 kg/m² and < 25 kg/m²), physical activity (inactive, moderate, vigorous), alcohol consumption (currently drinking or not), smoking status (current, former, and never smokers), and history of diabetes mellitus (yes/no), dyslipidemia (yes/no), and hypertension (yes/no).

between VOCs, airborne heavy metals, and brain MRI markers is scarce. All the VOCs assessed in the present study were inversely associated with whole-brain cortical thickness in the single-pollutant model. Among these VOCs, acetaldehyde was significantly associated with an increased AD-specific cortical atrophy score in the single-pollutant and mixture models. This epidemiological evidence on the neurotoxicity of exposure to VOCs is underpinned by experimental evidence. *In vitro* and *in vivo* studies have suggested that VOCs can function as neurotoxins, inducing oxidative stress and systemic inflammation (Kamel and Shehata, 2008; Mögel et al., 2011; Wang et al., 2014; Kodavanti et al., 2015; Wang et al., 2018).

Furthermore, an *in vitro* study revealed that exposure to styrene and toluene upregulates the expression of cyclooxygenase-2 (COX-2), an enzyme that regulates inflammatory responses (Mögel et al., 2011). Additionally, an animal study demonstrated that acetaldehyde exposure increases systemic inflammation by modifying self-proteins and inducing autoimmunity (Kamel and Shehata, 2008). This inflammatory process may damage the brain, resulting in human brain atrophy and cognitive impairment. Similarly, in this study, we observed an association between inhalation exposure to Mn₁₀ and reduced whole-brain cortical thickness. Although Mn is an essential mineral that is absorbed after ingestion, its long-term inhalation generally results in over-exposure (Miah et al., 2020). Moreover, Mn can circulate from the blood into the brain at high concentrations by crossing the blood-brain barrier

and accumulating in the brain cortex (Zheng et al., 1998). Additionally, Mn accumulation in the brain may damage the brain cortex by inducing neuroinflammation, oxidative stress, and dysregulation of neuronal transporters of Mn (Miah et al., 2020).

To the best of our knowledge, no neuroimaging study has considered multiple exposures to various air pollutants. In this study, we employed state-of-the-art approaches, including BKMR and QGcomp methods, to estimate the effects of air pollutant mixtures on subclinical imaging markers related to AD. The BKMR method flexibly selects variables from exposure vectors that exhibit high correlation and non-linearity (Bobb et al., 2018). The BKMR model demonstrated non-linear relationships between air pollutant mixtures and subclinical imaging markers in the present study. Specifically, we found a significantly lower AD-specific cortical atrophy score and a significantly greater whole-brain cortical thickness at the 25th percentile than at the 50th percentile of the air pollutant mixtures. However, the effects of the air pollutant mixtures remained similar between the 30th and 75th percentiles; this finding suggests that dwelling in areas with low-level concentrations of air pollutant mixtures (e.g., the 25th percentile in our cohort) may be beneficial for individuals without dementia in terms of preventing brain cortical atrophy. In our cohort, the 25th percentiles were 47.3 μm^3 for PM₁₀, 25.3 μm^3 for PM_{2.5}, 16.6 ppb for NO₂, 9.6 μm^3 for MN₁₀, 9.4 μm^3 for acetaldehyde, 12.7 μm^3 for styrene, and 27.8 μm^3 for toluene. These levels may need to be considered when revisiting air

quality guidelines to improve brain health. The QGcomp models showed that a one-quartile increase in air pollutant mixtures was significantly associated with a smaller whole-brain cortical thickness (0.30 mm). The positive association between air pollutant mixtures and the AD-specific cortical atrophy score did not reach statistical significance. However, this does not necessarily mean a null association because the QGcomp is a parametric method based on a linear model (Keil et al., 2020). Unexpectedly, toluene in the mixture showed a negative weight effect on the AD-specific cortical atrophy score, which was the opposite direction of the effect estimated from the single air pollutant model. This discrepancy may be due to the interactions of toluene with other highly correlated pollutants (e.g., Mn₁₀, acetaldehyde, and styrene). These possible interactions between pollutants may have amplified or attenuated the effect of toluene and changed the direction of its effects when it was present in the mixture (Schmidt, 2020; Letellier et al., 2022). Among the various pollutants studied, NO₂ substantially contributed to the effects of the air pollutant mixture on subclinical imaging markers. NO₂ is a highly reactive gas that promotes oxidative stress, resulting in the production of reactive oxygen species and subsequent mitochondrial dysfunction. These processes impair energy metabolism, reduce ATP production, and disrupt neuronal and vascular homeostasis, thereby increasing the risk of neuronal injury and cognitive decline (Yan et al., 2015). Furthermore, the harmful effects of NO₂ on subclinical imaging markers increased as the levels of the other pollutants decreased in the BKMR model; this suggests that individuals without dementia may be at risk of AD-like brain cortical atrophy due to exposure to traffic-related gaseous pollutants even when PM and other gaseous pollutants remain low, underscoring the importance of managing traffic emissions to reduce future dementia burden.

Although several studies have suggested that air pollution exposure deteriorates specific cognitive function domains (including executive functioning, immediate and delayed recall, and language), whether subclinical brain cortical atrophy mediated these effects is unclear (Wellenius et al., 2012; Tonne et al., 2014; Nußbaum et al., 2020). The present study identified, for the first time, the mediating roles of subclinical imaging markers in the effects of air pollutant mixtures on specific cognitive function domains using PLS-SEM. This analysis revealed that exposure to air pollutant mixtures was associated with decreased frontal/executive functions, language, visual memory, and visuospatial memory mediated by brain cortical atrophy in an AD-specific manner in adults without dementia. Notably, cortical atrophy in specific regions did not mediate the effects of air pollutant mixtures on their relevant cognitive function domains (data not shown). These findings align with the current neuroscience perspective, which emphasizes that the brain network, rather than a single brain region, is the most fundamental unit of neurobiological function (Petersen and Sporns, 2015). We speculate that subclinical cortical atrophy in multiple AD-related regions due to air pollution exposures may reflect brain structural network changes that can disrupt functional network dynamics (Jauny et al., 2024). Although a few studies have examined the effects of air pollution on brain functional networks (Glaubitz et al., 2022), future neuroimaging studies are required to investigate the effects of air pollution mixtures on structural and functional networks and examine the cognitive function domains impaired by these effects in the subclinical stage of AD.

However, this study has few limitations. First, the path we tested in the PLS-SEM analyses was based on cross-sectional data, in which brain atrophy (mediators) and specific cognitive function domains (outcomes) were measured simultaneously. However, the path is supported by sufficient evidence indicating that brain cortical atrophy in the subclinical stage precedes the onset of cognitive impairment and AD (Braak and Braak, 1991; Jack et al., 2010). Second, this study only considered seven pollutants as a mixture. However, this study is the first to estimate the combined effects of particulate and gaseous pollutants on brain structures and cognitive function. Further research is necessary to consider other gaseous pollutants, such as sulfur dioxide and ozone.

Third, we evaluated Mn₁₀ and VOC exposures using a spatial model derived from outdoor measurements collected from a subset of the study population between fall 2016 and summer 2017. The models for Mn₁₀ and VOCs demonstrated moderate performance; however, this exposure assessment method may have influenced our results. For example, Mn₁₀ and VOCs were strongly correlated; this may not have biased our results in the BKMR analyses, which can account for highly correlated air pollutants (Bobb et al., 2015). We also performed dimensionality reduction using the principal component analysis method in the PLS-SEM analyses. Fourth, the predictive performance of the exposure models varied across pollutants. Such variability, particularly for pollutants with lower predictive accuracy, may have introduced nondifferential measurement error, potentially biasing associations toward the null. Nevertheless, both BKMR and QGcomp analyses indicated that the relative contributions of individual exposures within the mixture to subclinical imaging markers were not driven by differences in model predictive performance. Finally, the present study included older Korean adults, which may limit its generalizability to other populations. However, sufficient evidence suggests that the effects of air pollution exposure on brain structures and cognitive function are also observed in other ethnic populations and geographical regions (Peters et al., 2015; Power et al., 2016).

5. Conclusion

In conclusion, the present study demonstrated that exposure to mixtures of particulate and gaseous pollutants was associated with brain cortical atrophy in adults without dementia. AD-specific cortical atrophy mediated the effects of these mixtures on visuospatial function, language function, visual memory, and frontal/executive function, indicating that air pollution may contribute to both preclinical and clinical phenotypes of AD.

These findings highlight the potential neurotoxic effects of ambient air pollution on brain health and the importance of implementing environmental interventions and comprehensive multipollutant management strategies. Reducing ambient air pollution may represent an important policy target for preserving brain health and mitigating dementia risk, while the identification of modifiable risk factors in future studies could support complementary intervention approaches.

CRedit authorship contribution statement

Heeseon Jang: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Hyunji Park:** Investigation, Data curation. **Young Noh:** Validation, Investigation. **Sang-Baek Koh:** Validation, Investigation. **Seung-Koo Lee:** Validation, Investigation. **Jung Il Lee:** Validation, Investigation, Data curation. **Ho Hyun Kim:** Validation, Investigation, Data curation. **Sun-Young Kim:** Validation, Investigation. **Jaelim Cho:** Writing – review & editing, Supervision, Conceptualization. **Changsoo Kim:** Writing – review & editing, Supervision, Conceptualization.

Data access, responsibility, and analysis

The datasets generated and/or analyzed during the current study are not publicly available to ensure privacy protection of the participants but are available from the corresponding authors on reasonable request.

Data statement

Anonymized data not published within this article will be made available by request from any qualified investigator.

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The funder had no role in the design and conduct of the study, the collection, management, analysis, and interpretation of the data, the preparation, review, or approval of the manuscript, or the decision to submit the manuscript for publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

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